

Environmental and economic appraisal of power generation capacity expansion plan in Nigeria

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ABSTRACT

The power sector in Nigeria is undergoing structural reforms aimed at improving and expanding the current grid generation capacity and distribution network. The Government has injected huge funds into this sector while also granting licences to private companies for the provision of electricity. It is also aiming to increase electricity generation capacity to 25,000 MW by 2020 from the current installed capacity of 6500 MW while also pledging to connect 75% of the population to the grid from the current 40% by 2025. This paper sets out to analyse the implications of the energy policy in Nigeria and presents the life cycle environmental and economic analysis of the current and future electricity sector. The results show that all the life cycle impacts and economic costs increase significantly over the time-period (2003–2030), but at different rates depending on the types of technologies deployed. Renewables such as large hydro and solar proposed by the Government have a potential to reduce the overall life cycle environmental impacts from the electricity mix, considering their lower environmental impacts compared to fossil-fuels. However, this requires a five-fold increase in grid investments from the current US\$1.7 billion per year to US\$9.40 billion by 2030.

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1. Introduction

An improved and efficient electricity supply system is one of the major socio-economic priorities for Nigeria. Despite electricity generation increasing by more than ten-fold from about 532 MW in 1972 to about 6500 MW in 2005 (Adenikinju, 2008), electricity supply is far below the demand, which is estimated at 10,000 MW (Tallapragada, 2009). The low development in this sector over the years has been largely due to neglect by the Government in terms of funding and capitalisation coupled with mismanagement and lack of maintenance (Ibitoye and Adenikinju, 2007). The Government's policy of sole ownership of electricity generation and distribution infrastructure in Nigeria has also starved this sector of private funding and market competition. This has translated into issues such as high costs of maintenance, low efficiency in generation, gas supply and revenue collection. Other reasons for the lack of progress in this sector include low water levels at the hydro-stations, vandalism of infrastructure, electricity theft and political instability (Obadote, 2009). Thus the performance of this sector over the years has been unsatisfactory.

In the early 2000s, transmission and distribution losses from the grid amounted to almost 40% (Ibitoye and Adenikinju, 2007)

and about 40% of the existing generation capacity was not in operation. For example, in 2003, only about 3800 MW of the generation capacity was available out of the installed capacity of 6500 MW. Moreover, with a population of over 140 million and with about 50% living in the rural areas (The World Bank Group, 2007), only 40% of the total Nigerian population have access to electricity (Babatunde and Shuaibu, 2009; Sambo, 2009; Mbendi, 2005) and only 10% of rural inhabitants are connected to the national grid (Mbendi, 2005; Obadote, 2009). More than a third of the power stations (36%) are over 20 years old, 48% are over 15 years old while about 10% are over 80 years old (Adenikinju, 2003).

In response to these challenges, the Government proposed and submitted the Electric Power Sector Reform Bill (EPSRB) to the National Assembly, which was passed in 2003 (Ikeme and Ebohon, 2005). This Bill established the Electric Power Sector Reform act replacing the previous Electricity Acts which recognise the Government as the sole electricity supplier and regulatory authority in the country. Under these reforms, many electricity generation and distribution licences have been granted to Independent Power Producers (IPPs) and Joint Venture Independent Power Producers (JVIPPs) in addition to increased investment in this sector by the Government. These reforms are already yielding results as several IPP and JVIPP companies have applied for or have been granted licences while several other power plant projects have been proposed by the Government in

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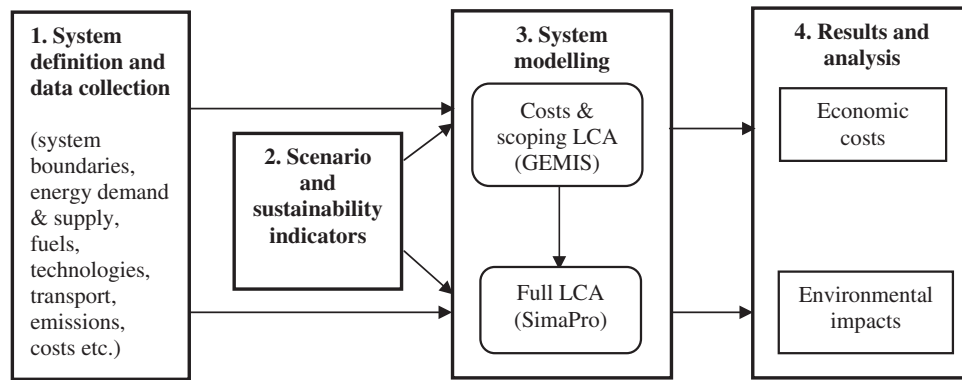


Fig. 1. Overview of the methodology used to evaluate the environmental and economic sustainability of the electricity system in Nigeria.

addition to several being built (Ibitoye and Adenikinju, 2007). The Government is also currently expanding its electricity programme to rural areas as part of its aim to extend access to communities that had been left out from previous electrification programmes.

With these on-going changes, the Government is aiming to increase power generation capacity from the current 6500 to 10,000 MW by 2010 (NNPC, 2005; Ukpohor, 2009) and over 25,000 MW by 2020 (ECN and UNDP, 2005), which is aimed at supplying electricity to about 75% of the population by 2025 (ECN, 2003). In its Renewable Energy Master Plan (REMP), the Government also plans to introduce more renewable energy systems into the electricity mix including solar, biomass, small hydro and wind (ECN and UNDP, 2005). These developments indicate that electricity generation in Nigeria is set to increase significantly over the time period. This will inevitably lead to disruptive changes in the energy system in terms of environmental, economic and social impacts, which should be assessed and evaluated in order to plan carefully and avoid potential pitfalls.

This study attempts to contribute towards such an evaluation and presents the results of an environmental and economic analysis of the Nigerian electricity mix, considering both the current and proposed power plants. The environmental analysis has been carried out using life cycle assessment (LCA). Several LCA studies of the energy system have been conducted that include comparison of different energy systems (Dones et al., 2004; Hondo, 2004; Kannan et al., 2006; Pieu, 2007; WEC, 2004); study of conventional energy systems (Gorokhov et al., 2002; Ruether et al., 2004); emerging systems such as fuel cells (Pehnt, 2003); renewable systems such as solar and wave energy systems (Banerjee et al., 2006), etc. LCA has also been used to study future energy systems in various countries (Koskela et al., 2007; Meier et al., 2005; Pehnt, 2006). As far as we are aware, this is the first time an LCA of the electricity options in Nigeria has been carried out. The following section outlines the methodology used for these purposes.

2. Methodology

The methodology used for evaluating the environmental and economic sustainability of energy systems is shown in Fig. 1. This methodology involves integrating LCA with economic costing to assess energy systems. The methodology involves defining the system boundaries and characteristics of the electricity systems and the collection of appropriate data for the study. The technical, environmental and economic data collected are then used to develop the current and future electricity scenarios to be used in

the modelling stage. The GEMIS 4.3¹ and SimaPro 6² softwares and databases have been used to model the system. The economic and environmental implications of the results obtained in the modelling stage are then analysed, interpreted and discussed in the final step of the methodology.

The following sections outline the methodological approach in more detail.

2.1. System definition and data collection

This analysis considers only electricity generation from the national grid, which comprises the national power plants and IPPs that supply power to the national grid. Due to data limitations, private electricity generation by individuals and businesses are not included in this analysis.

For the purposes of the analysis, the data collected are grouped into four major categories: primary, secondary, generic and scenario data. The primary data include data specific to Nigeria regarding installed capacity and generation of electricity, types of technologies, efficiencies, fuels costs resources data, etc. These were collected from ECN and UNDP (2005), EIA (2005b, 2006, 2007a, 2007b), NNPC (2004) and various literature sources including Akinbami (2001) and Akarakiri (1999). Secondary data have been calculated using primary and generic data and they include operation hours and fuel consumption. Generic data have been used where specific data for Nigeria have not been available and have been sourced from the GEMIS and SimaPro databases. Examples include emissions from power plants and fuels processing and production and processing of auxiliary materials such as steel and cement. These data have been modelled to represent Nigerian conditions, for example, using similar technologies, the efficiencies and operating hours of power plants in Nigeria have been used to estimate the emissions from the power plants. Fuels and auxiliary materials processing and production have also been modelled to represent Nigerian conditions. For example, associated-gas flaring of 70%³ has been used to account for crude oil production in Nigeria, in 2003. The scenario data include the

¹ GEMIS – Global Emission Model for Integrated Systems – Version 4.3 is an integrated LCA and economic analysis software and database for energy systems developed by the Öko Institute in Germany. More information Available at: <http://www.oeko.de/service/gemis/en/index.htm>.

² SimaPro 6 is an LCA software developed by Pré Consultants, The Netherlands. It contains the LCI of a wide range of materials and processes including energy and transport systems. More information Available at: <http://www.pre.nl/simapro.html>.

³ Nigeria flares in excess of 70–75% of the total associated gas produced (ERA, 2005; NNPC, 2005; Sonibare and Akeredolu, 2006).

Table 1

The base-year electricity mix (NEPA, 2000, 2004; Amioku, 2002; Akinbami, 2001).

Power station	Efficiency (%)	Installed capacity (MW)	Available capacity (MW)	Average operating time (2003) (h/a)	Status as of 2006
Gas_GT_Afam	18.93	710	568	2445	Working
Gas_GT_Delta	20.74	600	180	2445	Working
Gas_ST_Egbin	36.06	1320	528	2445	Working
Gas_GT_Ijora	31.55	600	480	2445	Working
Gas_GT_Sapele	20.74	300	240	2445	Working
Gas_ST_Sapele	27.95	720	576	2445	Working
Gas_GT_Lagos-Enron	30.65	270	216	2445	Working
Oil_ST_Abuja	36.96	30	24	2445	Working
Oil_ST_Calabar	38.27	2	2	2445	Working
Hydro_Jebba	35	560	280	8221	Working
Hydro_Kainji	31	760	228	8221	Working
Hydro-Shiroro	31	600	480	8221	Working
TOTAL		6472	3802		

Table 2

Future additions to the electricity generation mix in Nigeria (ECN and UNDP, 2005; Ibitoye and Adenikinju, 2007; Obadote, 2009; Okpanefe and Owolabi, 2002).

Technology type	Capacity (MW) 2010	Capacity (MW) 2020	Capacity (MW) 2030
Gas	6901	2284	1320
Coal	388	n/a	1320
Hydro (large)	n/a	4140	5748
Hydro (small)	n/a	600	n/a
Biomass	n/a	5	5
Wind	n/a	20	20
Solar PV	n/a	75	425
Solar thermal	n/a	1	20
Total addition	7289	7125	8858
Cumulative total	13,761	19,276	25,394

Government's plans for the expansion of the national grid in the future obtained from various sources.

2.2. Scenario development and sustainability indicators

This step involves the use of economic, environmental and technological data obtained from the previous step to develop the current and future energy scenario to be used in the next modelling stage.

2.2.1. Scenario development

Scenario analysis has been used to evaluate the environmental and economic implications of the proposed changes to the electricity sector in Nigeria. Four different time-periods have been considered: base-year (2003), 2010, 2020 and 2030.

2.2.1.1. Base-year (2003)—technical data. The electricity generation mix of the Nigerian grid consists of eight power stations connected to the national grid, and two power stations operated by IPPs, also connected to the grid. The installed capacity consists of 69.5% gas, 30% hydro and 0.5% diesel systems (Table 1). Due to limited data on the operating hours of these power plants, the total electricity generation and contribution of each technology type in 2003 has been used to determine the average operating hours for the power plants. The country generated 15.59 billion kWh of electricity in 2003 with hydropower accounting for about 59.1% of the total electricity generated and the rest being provided by the thermal power plants (EIA, 2005a). The lower amount of electricity generated from the thermal plants suggests that not all the units installed in the power plants were actually

producing electricity. For example, only about 2800 MW capacity was available for thermal power generation with installed generating capacity of over 4500 MW (Table 1). The idle units have been proposed for refurbishing and are considered in future years. The average efficiencies and capacity utilisation factors for the plants have been obtained from Akinbami (2001) and NEPA (2000, 2004).

2.2.1.2. Future years—technical data. The definition of the electricity mix for the years 2010, 2020 and 2030 has been guided by the Government's proposed and planned power plants for the respective years as well as the ones already built or being built. Various other renewable energy technologies – wind, biomass, small hydro and solar – planned or proposed by the Energy Commission of Nigeria, have also been used to define the future electricity mix (ECN and UNDP, 2005). The power plants are categorised into the estimated periods in which they will be available according to the Government's plans. For example, the plants that are already being constructed and the ones for which contracts have already been awarded are modelled for the year 2010.

(A) *Year 2010*⁴: It is planned that the additional generation capacity by 2010 of 7289 MW will mainly comprise conventional systems made up of 95% gas and 5% coal as shown in Table 2. This will be 'business as usual', mainly because fossil fuels play a key role in electricity generation. All the power plants in the base-year are expected to be in operation in 2010, thus resulting in a cumulative installed capacity of 13,761 MW consisting of 83% gas, 13.9% hydro, 2.8% coal and a minute contribution from diesel (0.3%). However, only the available capacity in the base-year have been used for modelling since most of the units in the existing power plants are very old, as mentioned earlier.

(B) *Year 2020*: A further addition of about 7125 MW power plants to the grid will have been achieved by 2020. This includes the ECN's 2015 target for the renewable energy (ECN, 2005), which would consist primarily of hydropower (66.5%) and gas (32%) with new renewable energy technologies accounting for the rest (see Table 2). These include biomass (5 MW), solar PV cells (75 MW), solar thermal (1 MW) and wind (20 MW). A key assumption for this scenario is that some of the power plants in the base-year will be decommissioned by 2020. For this analysis, the dates of construction and extensions to these power plants have been used. Thus the Afam (710 MW), Ijora (600 MW) and Sapele-ST

⁴ At the time of writing, the 2010 scenario is far from the target of at least 10,000 MW generation capacity planned by the Government due to on-going unrest in the Niger Delta region and on-going construction of gas pipelines. The Government has however reiterated its position to achieving this target by end of 2011 (AfDB, 2009).

(300 MW) power plants will have been outdated even though some of the units will still be within their lifetimes. This is evened out by the assumption that the remaining plants will have been rehabilitated to operate at higher capacities. Thus, a cumulative capacity of 19,276 MW is assumed for 2020.

(C) *Year 2030*: An additional 8858 MW installed capacity consisting of 70% renewables and 30% conventional systems will be added to the grid between 2020 and 2030. By 2030, biomass and wind energy technologies will still be on a small-scale while solar technologies (mostly PV) will have increased by almost seven times from 76 MW in 2020 to 521 MW by 2030 according to the Government's plans. It is also expected that a number of the plants in the base-year will not be operational by 2030. Thus, using the same assumptions deployed in 2020, the only power plants expected to be operating by 2030 are the two IPPs (300 MW) and the hydropower plants. The cumulative installed generation capacity will have reached 25,394 MW by 2030, indicating an increase of almost four times compared to the base-year. This also suggests an annual growth rate of about 5% in generation capacity over the period (2003–2030).

(D) *General assumptions*: The efficiencies, annual operating hours and lifetimes of the future power plants have been obtained from literature sources. The efficiencies and life times for solar PV, wind and solar thermal plants represent the electrical efficiencies as obtained from the LCA software GEMIS (Oko Institute, 2006). The average operating hours for wind power, solar PV and solar thermal are also sourced from GEMIS (see Table 3). The planned expansion of the transmission and distribution (T&D) network to connect 75% of the population by 2020 has also been taken into account in the modelling. Thus it has been assumed that the T&D network must at least double over the period.

2.2.1.3. Base-year—economic costs. All the costs are presented in 2005 US\$ using the consumer price index (CPI). The same approach has been followed for the future scenarios. As shown in Table 4, there is a considerable difference between domestic and international prices of fuels, which is due to low crude oil prices in Nigeria. The analysis here is based on the domestic prices of fuels, except for imported products (in this case

Table 3
Characteristics of future energy systems.

Technology	Efficiency (%)	Lifetime (yr)	Operating hours (hr/yr)
Gas	33.5	30	8500
Coal	38.25	30	8500
Hydro	35	80	8500
Solar-PV	100	20	3000
Biomass	35	30	8000
Wind	100	20	3600
Solar-thermal	100	25	3000

Table 4
Costs of fuels in Nigeria in 2003 (from various sources).

Fuels	Prices	References	Prices (US\$2005 per tonne)
Natural gas	\$116.10 per 1000 m ³	EIA (2003) ^a	178
Diesel (International)	N43.50 per litre	OPEC (2003)	419
Diesel (Domestic)	N32.50 per litre	NNPC (2004)	313
Fuel wood	N3.00 per kg	ECN (2004)	24
Coal	\$40.00 per tonne	NCC (2002)	42

The conversion rate for the Nigerian Naira (N) for 2003: N130=US\$1 (CBN, 2003).

^a Price conversion: 1 m³ of natural gas=35.31 ft³=38.3 MJ. Price converted to tonne from heating value of natural gas (49 MJ/kg) (Oko Institute, 2006).

Table 5
Technology costs.

Technology	Capital investment cost (US\$2005 per kW)	Fixed cost (US\$2005 per kW)	Variable cost (US\$2005 per kWh)	Reference
Gas turbine	607	11	0.003	Akinbami (2001); EIA (2005b)
Steam turbine	1289	11	0.005	Akinbami (2001); EIA, (2005b)
Hydro	2729	13	0.0005	Akinbami (2001); EIA (2005b)

Table 6
Future projections for fuel prices.

Fuels	Prices 2010 (US\$2005 per tonne)	Prices 2020 (US\$2005 per tonne)	Prices 2030 (US\$2005 per tonne)	Annual price growth rate (%) (EIA, 2006)
Natural gas	191	211	233	1.1
Diesel (International)	475	567	678	1.8
Diesel (Domestic)	355	424	507	1.8
Biomass (fuel wood)	25	25	25	0.1
Coal	42	43	43	0.1

diesel). The cost of diesel has been modelled at local prices for the locally produced portion and vice versa for the imported portion. It has also been assumed that there will be enough refining capacity to meet the demand in 2020 and 2030 due to the various refineries already being built in Nigeria; thus projected local prices are used in modelling the future energy mix. The costs of existing power plants have also been used in the analysis (Table 5).

2.2.1.4. Future years—economic costs. The costs of fuels for 2010, 2020 and 2030 have been projected from growth rates published by EIA (2006), as shown in Table 6. The price growth rates for biomass have been assumed to be the same as for coal (0.1%); this assumption is made due to the biomass prices being mostly independent of market forces in rural areas in Nigeria while also increasing slowly in the urban areas. The domestic prices of these fuels have been used to model the economic costs, except for imported petroleum products in 2003 and 2010 as mentioned in the previous section. The future costs of the new power plants and technologies have been obtained from the EIA (2007a, 2007b) as shown in Table 7.

2.2.2. Sustainability indicators

Two types of sustainability indicators have been used for the purposes of the analysis here: environmental and economic. The environmental indicators are defined by the environmental impacts used in life cycle assessment (LCA) and the economic indicators are those used in capital project evaluations. The LCA impacts have been calculated using the CML 2001 impacts characterisation method (Guinée et al., 2001) and comprise global warming (GWP), abiotic depletion (ADP), Ozone depletion (ODP), human toxicity (HTP), freshwater toxicity (FTP), marine toxicity (MTP), terrestrial toxicity (TTP), photochemical oxidation (POCP), acidification (AP) and eutrophication (EP).

The economic costs represent the total cost comprising annualised capital cost, annual fixed cost, annual variable cost

and annual fuel costs. The total cost has been calculated as follows:

$$T_C = \sum C_C + \sum F_C + \sum V_C + \sum f_C \quad (\text{US\$}/\text{yr}) \quad (1)$$

where T_C is the total annual cost; C_C the annualised capital cost; F_C the annual fixed costs; V_C the annual variable costs; and f_C the annual fuel costs.

The annualised capital cost (C_C) is calculated by multiplying the capital cost by an annuity factor (f) defined as

$$f = zx(1+z)^t / [(1+z)^t - 1] \quad (2)$$

where z is the discount rate (assumed at 13%) and t the lifetime of technology (yr).

Table 7
Future prices of technologies (EIA, 2007a,b; ECN, 2004).

Technology type	Investment cost (US\$2005 per kW)	Fixed costs (US\$2005 per kW)	Variable costs (US\$2005 per kWh)
Gas turbine 2010	594	11	0.003
Gas turbine 2020	576	11	0.003
Gas turbine 2030	561	11	0.003
Steam turbine 2020	405	11	0.005
Steam turbine 2030	399	11	0.005
Coal 2010	830	8	0.004
Coal 2030	800	8	0.004
Biomass 2020	1721	50	0.008
Biomass 2030	1534	50	0.008
Solar thermal 2020	2309	53	–
Solar thermal 2030	2067	53	–
Solar PV 2020	3569	11	–
Solar PV 2030	3923	11	–
Large hydro 2020	1500	13	0.0005
Large hydro 2030	1426	13	0.0005
Small hydro 2020	1980	12	–
Wind 2020	1194	29	–
Wind 2030	1194	29	–

Annual fixed costs reflect the costs to operate and maintain a process per unit capacity per year while the variable costs reflect current expenses of delivering a product or service per unit of energy per year excluding the fuel costs. The 'Unit' costs, which are calculated as the total cost divided by the total energy demand, have also been estimated.

2.3. System modelling

Two inter-linked models have been created to enable system modelling—one using GEMIS 4.3 and the other using SimaPro 6. Although both are LCA software packages, they have been used for different but related purposes. GEMIS has been used for the preliminary (scoping) LCA but also to calculate the economic implications of the current and future electricity mix. The preliminary results obtained by GEMIS have then been used in the SimaPro model to carry out full environmental impacts analysis. It was also necessary to use GEMIS first because, in addition to the scoping LCA and costs analysis, it also provided secondary data for use in SimaPro; for example final energy generation data using the properties of the various processes and fuels.

3. Results and analysis

3.1. LCA of the electricity sector

This section presents the LCA results for base-year and future electricity mix.

3.1.1. System boundary

The system boundary is from 'cradle to grave'; comprising the extraction and processing of fuels, transport of the fuels to the power plants and electricity generation from the power plants (see Fig. 2). This analysis also takes into account the building of infrastructure including power plants, refineries, oil rigs, pipelines and transmission and distribution equipment.

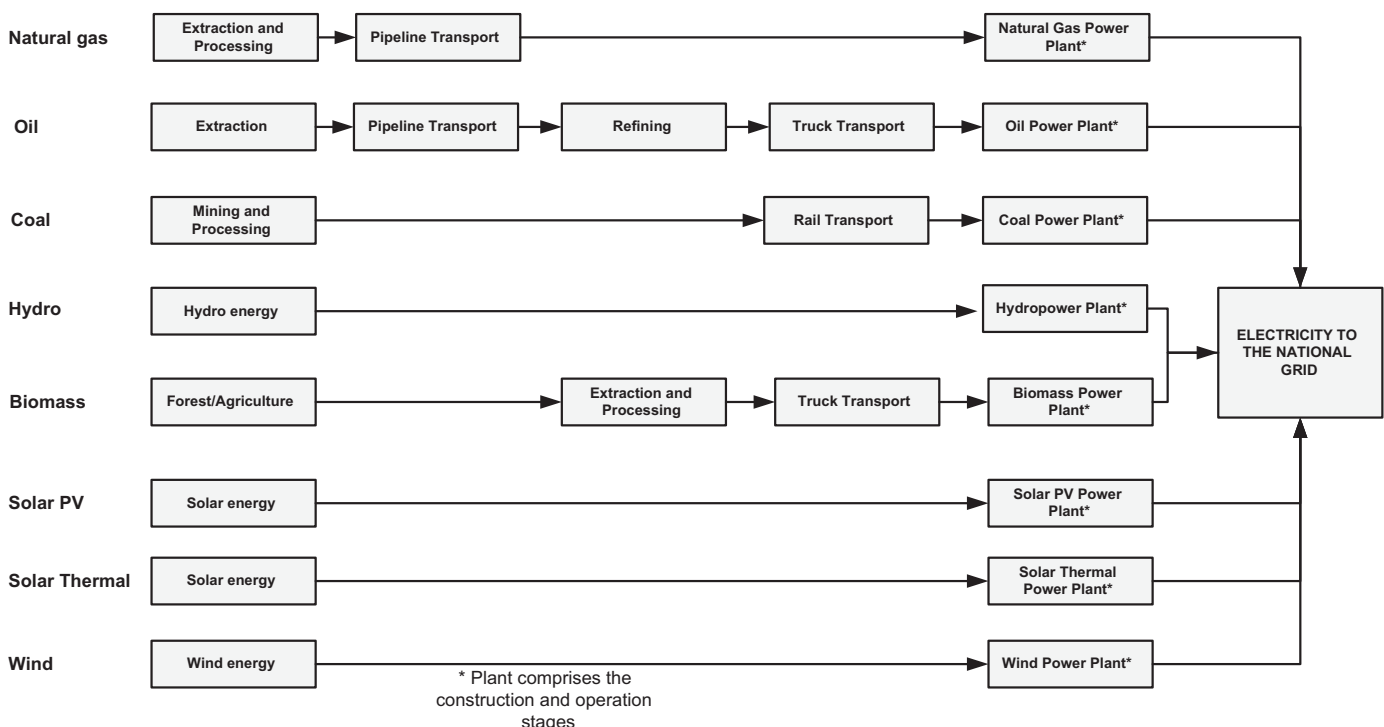


Fig. 2. Life cycle stages of the electricity system.

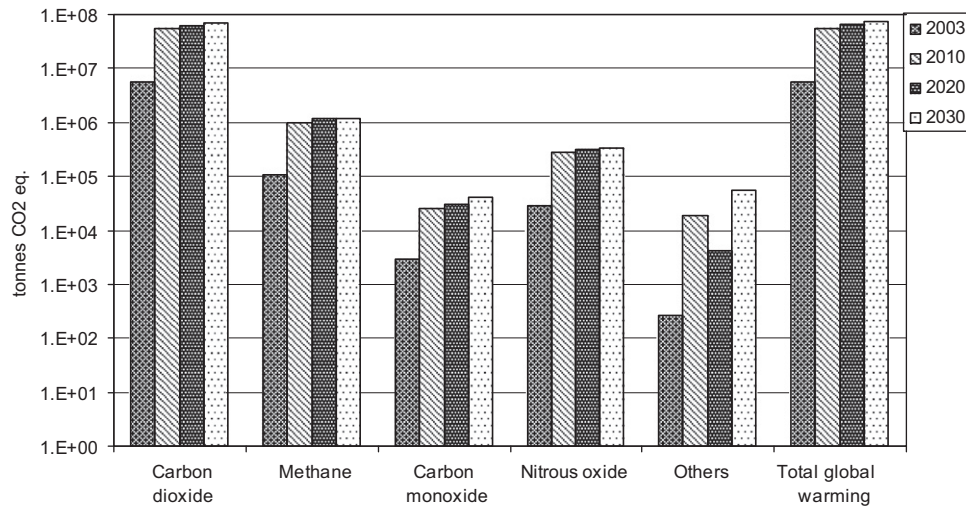


Fig. 3. Global warming potential for the electricity mix.

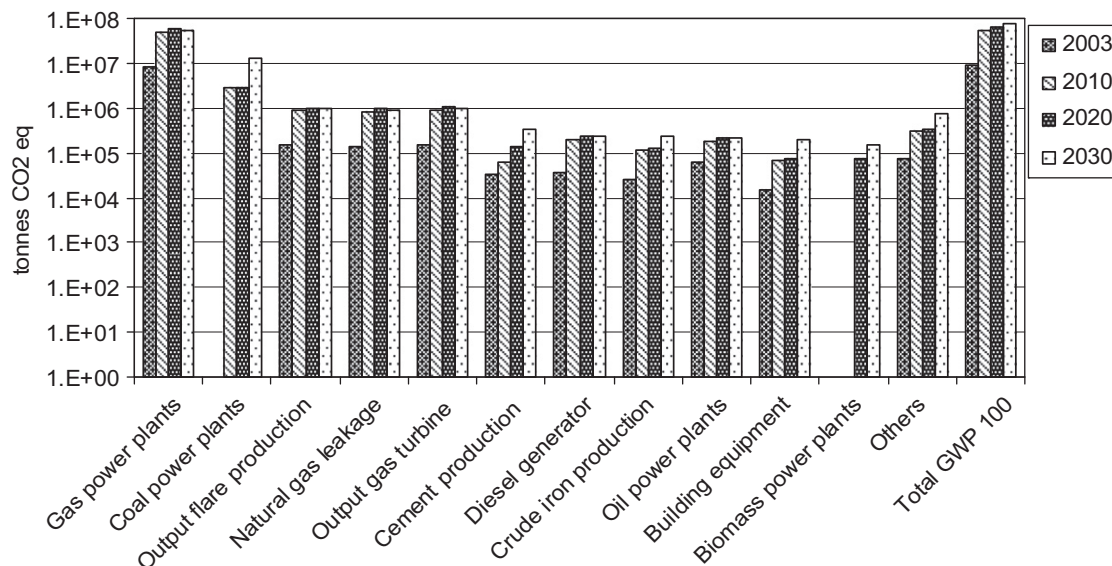


Fig. 4. Contributions to GWP100 for the electricity mix.

3.1.2. Unit of analysis (functional unit)

The environmental impacts are analysed based on the total electricity produced in a given year, that is:

- 56,160 TJ/yr (15.6 million MWh/yr) for 2003;
- 346,000 TJ/yr (96.1 million MWh/yr) for 2010;
- 551,000 TJ/yr (153.1 million MWh/yr) for 2020 and
- 764,000 TJ/yr (212.2 million MWh/yr) for 2030.

The above figures take into account the current or projected plant efficiencies, availability and operating hours, as appropriate for each time period.

3.1.3. LCA results

The LCA results are shown in Figs. 3–13. As shown, all the impacts increase significantly within the period due to increase in electricity generation and consumption but at different rates depending on the types of technologies and energy mix deployed over the time period. This is discussed in the following sections.

3.1.3.1. Global warming potential (GWP100). The total GWP100 (global warming potential over 100 years) increases by almost 13 times to 73.4 million tonnes CO₂ eq. by 2030 compared to about 5.7 million tonnes CO₂ eq. in 2003 (Fig. 3). Carbon dioxide accounts for over 97% of the total impact in all the years considered with methane, nitrous oxide and carbon monoxide accounting for the rest. The emissions from the combustion of natural gas and coal are the main contributors to the total GWP100 (see Fig. 4). The natural gas power plants account on average for 87% of the total GWP in 2003, 2010 and 2020. In 2030 this contribution reduces to 76% due to the reduced use of gas, and the contribution from coal power plants rises to 23% of the total GWP100 in 2030 from an average contribution of 5% in 2010 and 2020.

3.1.3.2. Abiotic depletion potential (ADP). The total ADP increases by 13 times over the period from just under 52,000 tonnes Sb eq. in 2003 to about 674,000 tonnes Sb eq. in 2030. Most of the total ADP occurs at the resource extraction stages that include natural gas production, coal mining, and gas processing (see Fig. 5). The contributions from these stages therefore follow the trend of consumption of these resources. Natural gas is the most

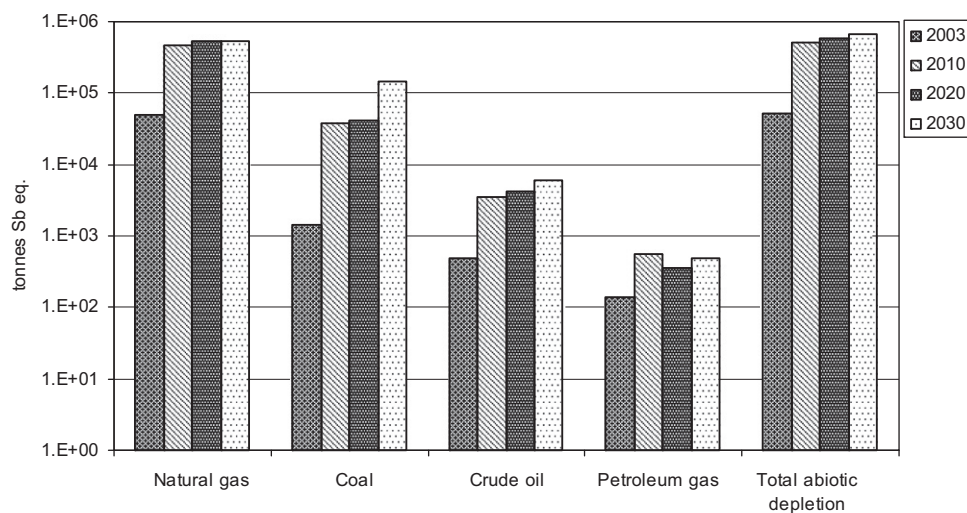


Fig. 5. Abiotic depletion potential for the electricity mix.

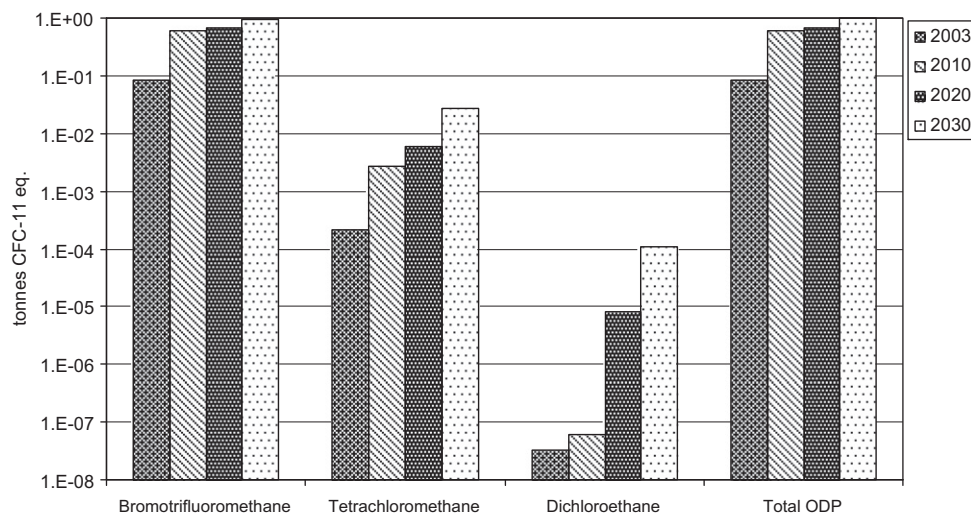


Fig. 6. Ozone depletion potential for the electricity mix.

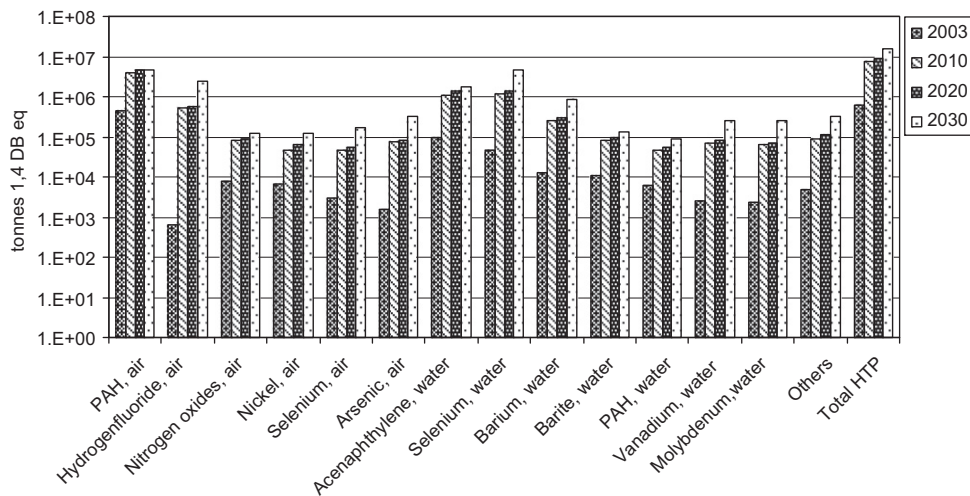


Fig. 7. Human toxicity potential for the electricity mix.

significant burden over the time period mainly due to the number of gas power plants, contributing to over 90% of the total ADP in 2003, 2010 and 2020 and over 70% contribution in 2030. The

relatively lower contribution in 2030 is due to the decrease in consumption of gas power in the electricity generation mix. The contribution from coal to total ADP increases from just below 3%

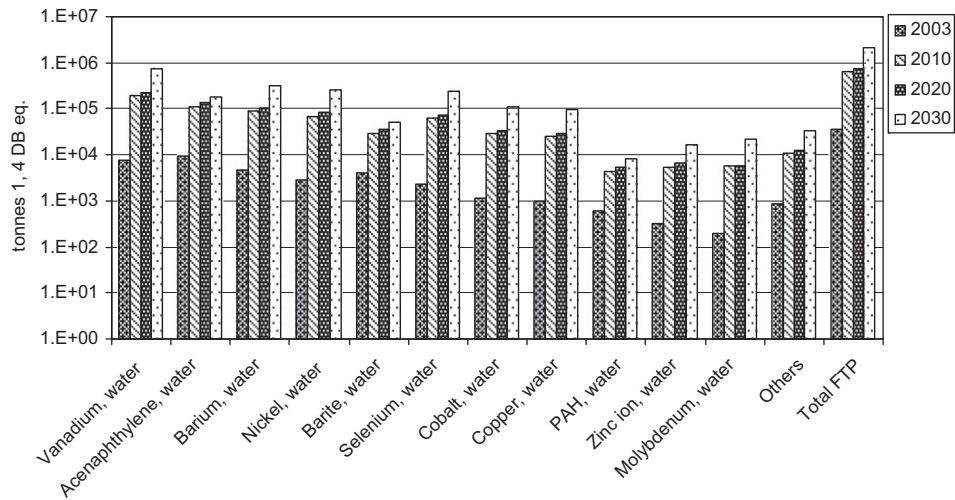


Fig. 8. Freshwater toxicity potential for the electricity mix.

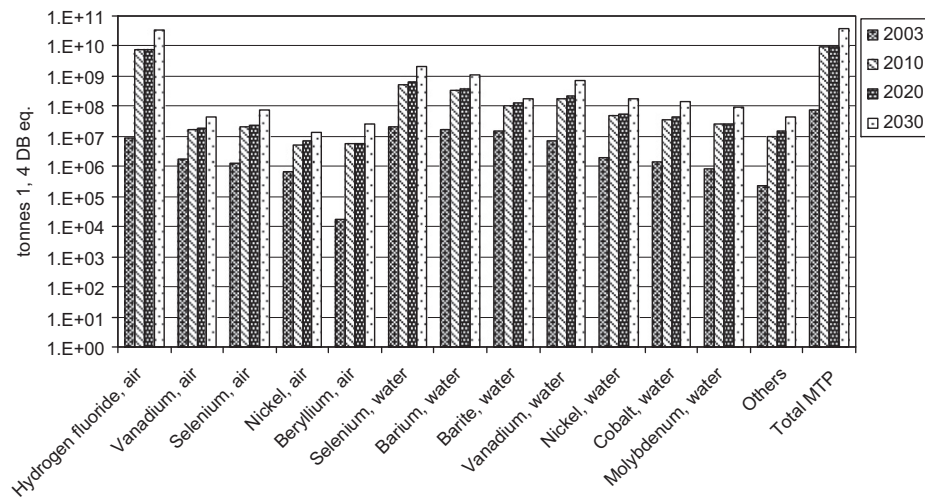


Fig. 9. Marine toxicity potential for the electricity mix.

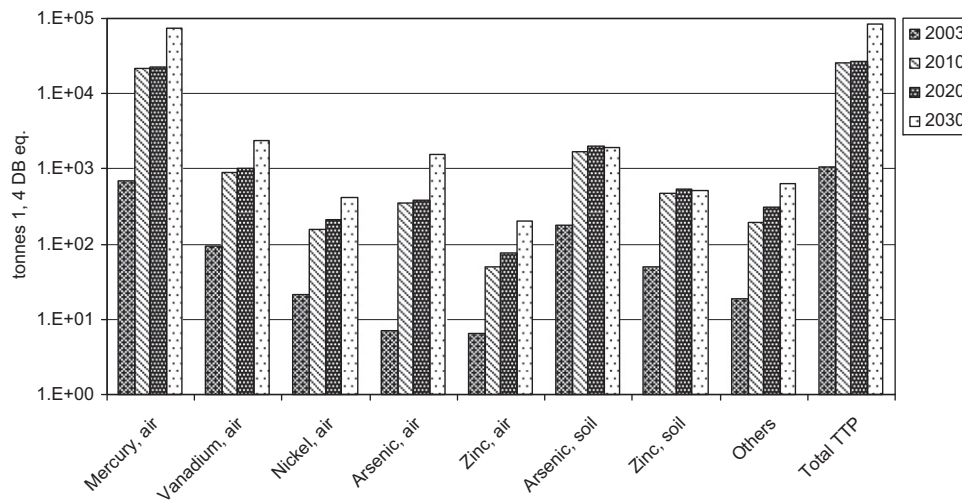


Fig. 10. Terrestrial toxicity potential for the electricity mix.

in 2003 to an average 7% in 2010 and 2020 due to the introduction of a 388 MW coal-fired power plant in 2010, where its contribution peaks at over 20% in 2030. This is as a result of additional

1320 MW coal-fired plants planned to be installed by 2030. There are only two oil power plants in the time-period with a combined capacity of 32 MW; thus the contribution of crude oil to the total

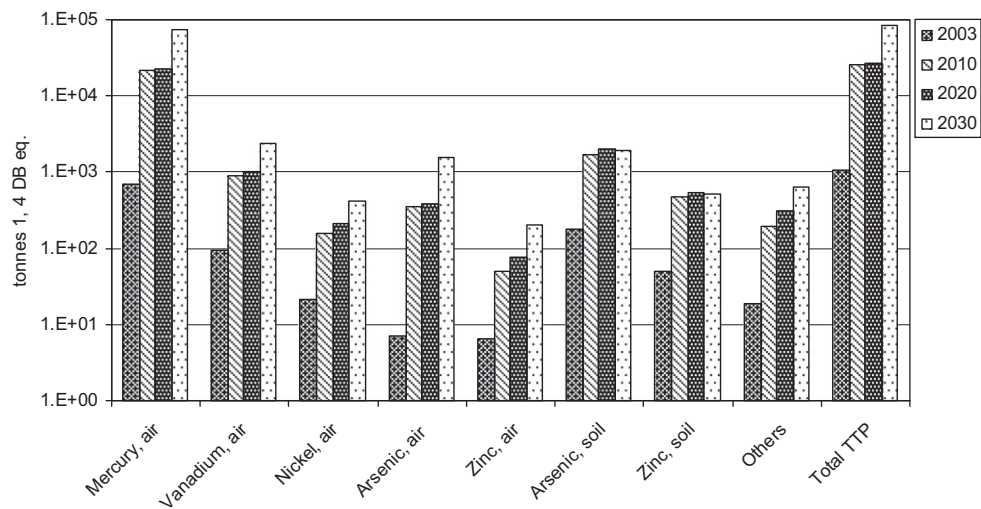


Fig. 11. Photochemical oxidation potential for the electricity mix.

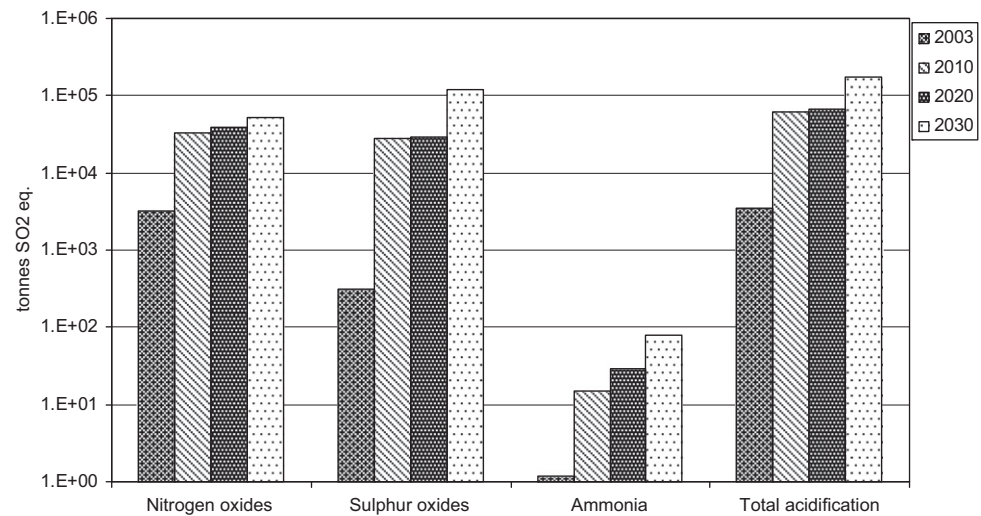


Fig. 12. Acidification potential for the electricity mix.

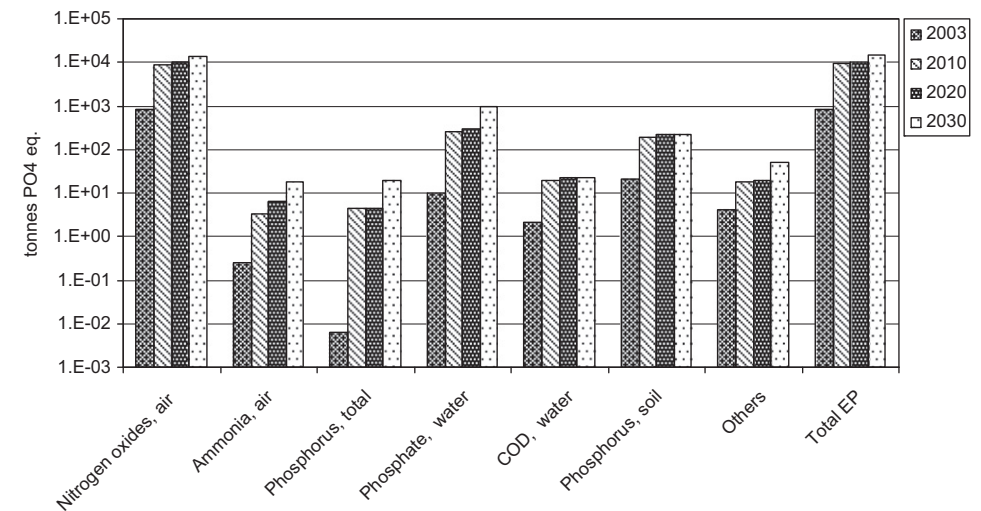


Fig. 13. Eutrophication potential for the electricity mix.

ADP is minimal, averaging less than 1% in all the years considered. Petroleum gas⁵ has an insignificant contribution to the total ADP in all the years considered (mainly due to gas flaring at the crude-oil production stage).

3.1.3.3. Ozone layer depletion (ODP). The total ODP also increases 12-fold by 2030 compared to the ODP of about 82 kg CFC-11 eq. in 2003 (see Fig. 6). Bromotri-fluoro-methane is responsible for most of the ODP constituting on average about 99% to the total ODP. Most of this burden occurs at the crude oil production stage. Thus crude oil production accounts for over 95% of the total ODP. There is also on average a 4.7% contribution from oceanic tanker transport in 2003 and 2010 due to petroleum products imports. This contribution reduces significantly in the subsequent years as petroleum products imports are no longer needed in the system.

3.1.3.4. Human toxicity potential (HTP). The total HTP peaks to about 16 million tonnes 1,4 DB eq. by 2030, representing an over 25-fold increase compared to the base-year (see Fig. 7). The emissions of poly-aromatic hydrocarbons (PAH) to air, selenium and acenaphthylene emissions to water and hydrogen fluoride emissions to air are the major burdens contributing to the total HTP. These are discussed below.⁶ PAH emissions to air increase six-fold over the time period. Although this burden increases over the period, its overall contribution to the total HTP reduces. For example, this burden constitutes 67% of the total HTP in the base-year while its contribution decreases to an average 52% in 2010 and 2020 and to just about 29% of the total HTP in 2030. There are two reasons for this: the reduced gas consumption by 2030 as the emissions from the gas power plants account for about 98% of the PAH emissions to air; and the increase in selenium emissions to air from coal power. Increased coal power consumption increases the contribution to the total HTP from selenium emission to water to 28% by 2030 from an average contribution of 7% in 2003. This is because coal tailing landfill accounts for over 97% of selenium emissions to water. There is also a contribution of 15% from hydrogen fluoride emissions to air by 2030, which is mainly caused by coal power plants.

3.1.3.5. Freshwater aquatic eco-toxicity potential (FTP). The total FTP shows a significant increase of almost 60 times to about 2 million tonnes 1,4 DB eq. in 2030 compared to the base-year. The major burdens contributing to the total FTP include emissions to water of vanadium, acenaphthylene and barium (see Fig. 8). The contribution of vanadium emissions in water to the total FTP averages 21% in the base-year, 30% in 2010 and 2020 and 35% in 2030. This trend follows the consumption of coal as coal tailings are responsible for over 97% of the vanadium emissions to water. This burden contributes most to the total FTP in all the years considered except in the base-year where acenaphthylene emissions to water contribute 28% to the total FTP. The contribution of this burden to the total FTP averages 17% in 2010 and 2020 to as low as 8.5% in 2030. This trend is attributable to the increases in other burdens from the power plants as most (99%) of the acenaphthylene emissions to water come from LDPE production, which is used for T&D lines, etc.

3.1.3.6. Marine toxicity potential (MTP). Of all the impacts, MTP shows the most significant increase in the time period considered. By 2030, it increases by more than 500 times to 38.9 billion

tonnes 1,4 DB eq. compared to 76.5 million tonnes 1,4 DB eq. in 2003 (see Fig. 9). Hydrogen fluoride emission to air, mainly from the coal power plants accounts for an average of 86% of the total MTP in 2010, 2020 and 2030. Selenium, barium and barite emissions to water are the major burdens contributing to the total MTP in 2003 with contributions of 27%, 22% and 19%, respectively.

3.1.3.7. Terrestrial toxicity potential (TTP). The total TTP increases by over 75 times over the whole period, mainly contributed to by mercury emissions to air. Mercury emissions to air constitute about 64% of the total TTP in the base-year; this contribution increases to 84% in 2010 and 2020 and 91% in 2030 (see Fig. 10). This burden is mainly from natural gas production and consumption activities and thus increases with increased gas consumption over the time period.

3.1.3.8. Photochemical oxidation potential (POCP). The total POCP increases 27 times over the time period to about 8000 tonnes C₂H₄ eq. in 2030. This impact follows the trend of coal and natural gas power consumption. The main burdens contributing to the total POCP are SO_x, ethane, pentane, carbon monoxide and methane in all the years considered (see Fig. 11).

3.1.3.9. Acidification potential (AP). There is a significant increase (almost 50-fold) in the total AP over the time period from about 3500 tonnes SO₂ eq. in 2003 to about 171,000 tonnes SO₂ eq. in 2030. This dramatic increase in AP is noticed as the emissions of SO_x increase due to increased coal power consumption in 2030. NO_x emissions mainly from natural gas power plants accounts for about 90% of the total AP in 2003. This contribution averages about 60% in 2010 and 2020 while decreasing to about 30% in 2030 due to the dramatic increases in SO_x emissions from increased coal consumption (see Fig. 12).

3.1.3.10. Eutrophication potential (EP). The total EP increases by over 17 times from about 860 tonnes PO₄ eq. in 2003 to 14,900 tonnes PO₄ eq. in 2030 (see Fig. 13). NO_x emission accounts for about 96% of the total EP in all the years considered. Emissions from the gas power plants contribute most to the total EP, accounting for 71% of the total in 2003, an average 62% in 2010 and 2020 and 43% of the total EP in 2030. However, increased coal power generation by 2030 also leads to a significant contribution of 30% to this impact from the coal power plants by 2030.

3.2. Economic analysis

The economic analysis has been carried out using a discounting rate of 13% throughout (CBN, 2006).⁷

3.2.1. Capital costs

The increased generation capacity from just under 6500 MW in 2003 to over 25,000 MW by 2030 increases the total capital costs by over three times from about US\$9.4 billion in 2003 to about US\$30.1 billion in 2030 (see Fig. 14). By 2010, the addition of 7289 MW of predominantly gas power plants to the grid increases the total capital cost by over 47% to US\$13.8 billion. The cost increases further by 57% in 2020 due to the addition of an extra 7125 MW to the grid where an additional 38% increase is expected in 2030. By then, an extra 8858 MW will have been added to the grid.

⁵ The system has been modelled with 70% gas-flare in the base-year.

⁶ Note that only the most significant burdens in terms of contribution to the total impacts are discussed.

⁷ The minimum discount rate dropped from 20.5% in 2001 to 15% in 2003 and 2004 and then to 13% in 2005 and first quarter of 2006. Thus the discount rate of 13% has been used in all the costing calculations.

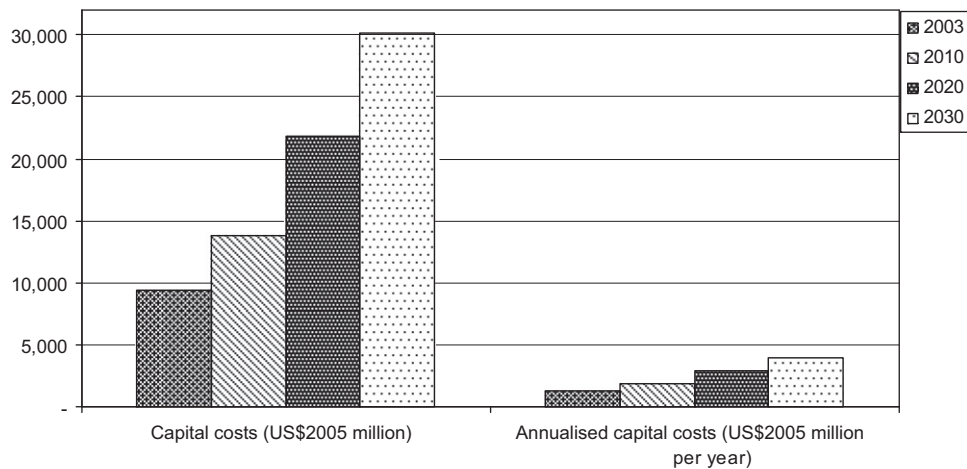


Fig. 14. Capital and annualised capital costs of the electricity mix.

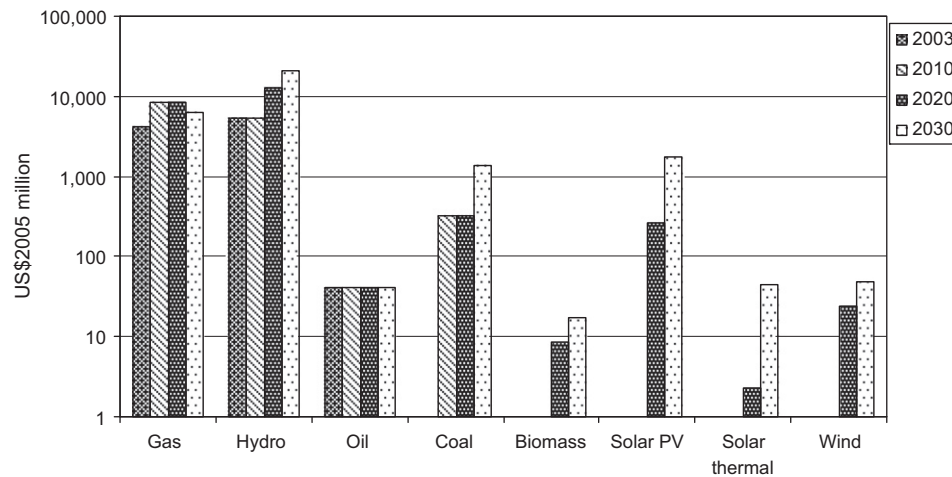


Fig. 15. Total capital costs by fuel type for the electricity mix.

Annualised capital costs: The total annualised capital costs follow the same trend as the total capital costs as discussed above, since they are calculated by multiplying the total capital costs with the annuity factor (Eq. (2)).

As shown in Fig. 15, hydropower contributes the highest to the capital costs in all the years considered due to the installed capacity as well as the higher costs of investment compared to the fossil-fuelled systems. The total capital cost for hydropower is expected to be in excess of US\$20 billion by 2030 from about US\$5.2 billion in the base-year, an increase of almost four times over the period reflecting a six-fold increase in the installed capacity from 1938 MW in 2003 to 12,308 MW by 2030. The contribution of the gas power plants to the total capital costs peaks in 2020 to about US\$8.5 billion from US\$4.1 billion in 2003 and then decreases to about US\$6.2 billion in 2030; this is due to the decrease in gas power consumption in the period.

3.2.2. Fixed, variable and fuels costs

The annualised fixed, variable and fuels costs are shown in Fig. 16. The total annualised fixed cost doubles to over US\$155 million per year in 2010 compared to US\$76 million per year in 2003. It then increases further by 46% to just under US\$230 million per year in

2020 with a further increase of 33% estimated for the year 2030. Electricity from natural gas contributes the most to the total fixed costs in 2003, 2010 and 2020, constituting 66%, 81% and 60%, respectively of the total fixed costs. The reduced cost contribution of the gas power plants in 2020 as well as in 2030 is due to the increase in the hydropower capacity, which with 53% has the highest contribution to the total fixed cost in 2030.

The variable costs exclude the fuel costs, which are calculated separately. The annual variable costs increase from about US\$51 million per year in 2003 to almost US\$400 million per year in 2030, an almost eight-fold increase. The gas power plants contribute the most to the total variable costs constituting 83%, 91%, 88% and 72% in 2003, 2010, 2020 and 2030, respectively. Renewable energy systems have little or no variable costs and thus do not contribute much to the total variable cost. For example, despite hydropower accounting for over 49% of the electricity generation in 2030, its contribution to the total variable costs is about US\$50 million per year, which is 17% less than the contribution from coal power, which accounts for just 6.8% of total electricity generation in 2030.

The fuel costs also increase by about 14 times from US\$333 million per year in 2003 to about US\$4.7 billion per year in 2030 as more fossil-fuel systems are introduced into the system. On

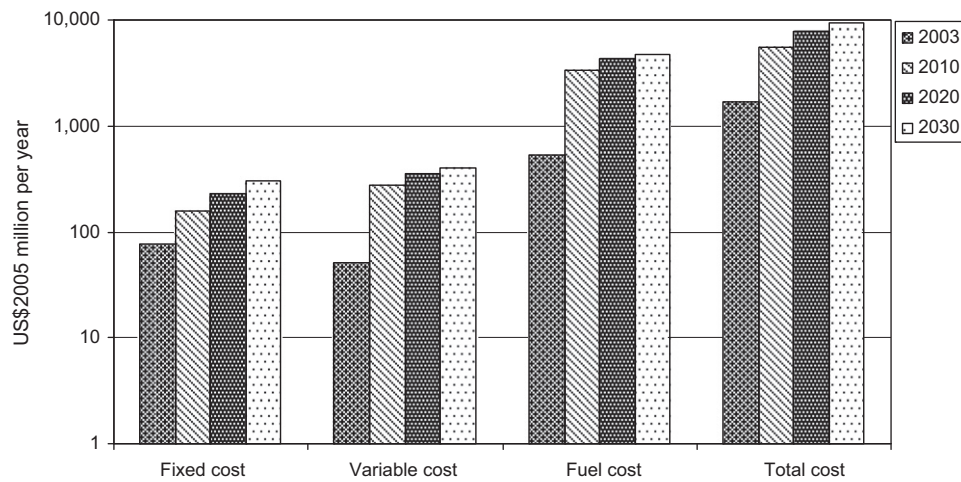


Fig. 16. Different costs for the electricity mix.

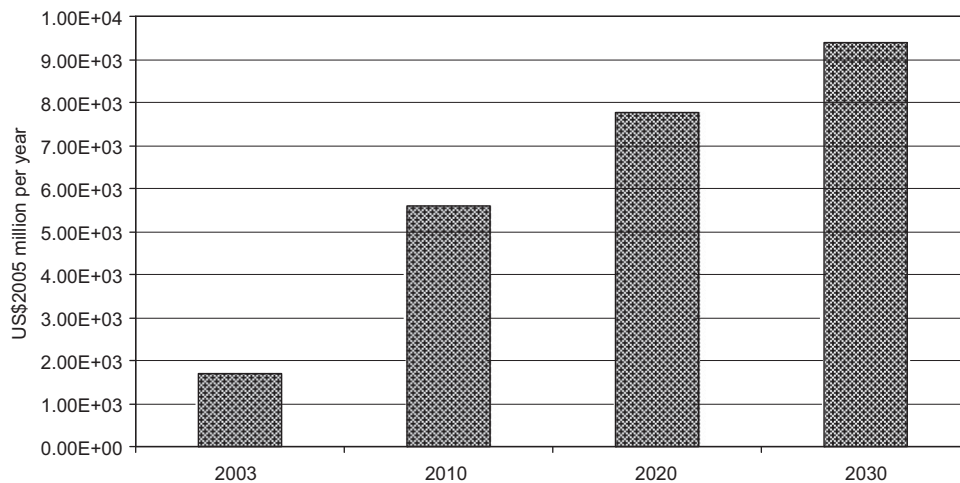


Fig. 17. Total costs for the electricity mix.

average, gas power accounts for over 95% of the total fuel costs in all the years considered as the projected consumption and cost of gas increase overtime.

3.2.3. The total cost

As shown in Fig. 17, the total cost is set to increase from US\$1.7 billion per year in the base-year to about US\$9.4 billion per year by 2030, an over five-fold increase over the period. The trend in the total costs suggests the highest increase between 2003 and 2010 because all the additional power plants will be fossil-fuelled (97% gas) implying higher fixed, variable and fuel costs (see Fig. 18). The total cost increase in the subsequent years is due to the addition of renewable power systems that may have little or no variable as well as no fuel costs, but are associated with higher capital costs.

3.2.4. Unit costs

The unit costs are set to decrease with time (see Fig. 19). The unit cost for the base-year is estimated at about US\$94 per MWh per year, which is set to decrease to about US\$44 per MWh per year by 2030; this is expected since the total energy generation grows

faster than the total costs. This implies it is will be cheaper to generate electricity in the future years compared to the base year.

4. Discussion

The on-going reforms in the electricity sector and proposed plans by the Government suggest electricity generation capacity is expected to increase by over four times over the study period (2003–2030). Moreover, the national grid is expected to connect well over 75% of the population compared to the current 40%. As a consequence, all the life cycle impacts will increase significantly over the time period, from about 13 times for GWP100 to 500 times increase in marine toxicity by 2030. Despite the large increase in the life cycle impacts, there are number of arguments that support the Government's initiative to expand the power sector. These include improving energy supply and security to maintain the natural growth in consumption, economic and social development priorities that call for an expanded access to electrification, addressing the impact from private generators, and irrigation and water supply benefits that could be derived from hydropower dams. It should also be noted that the same

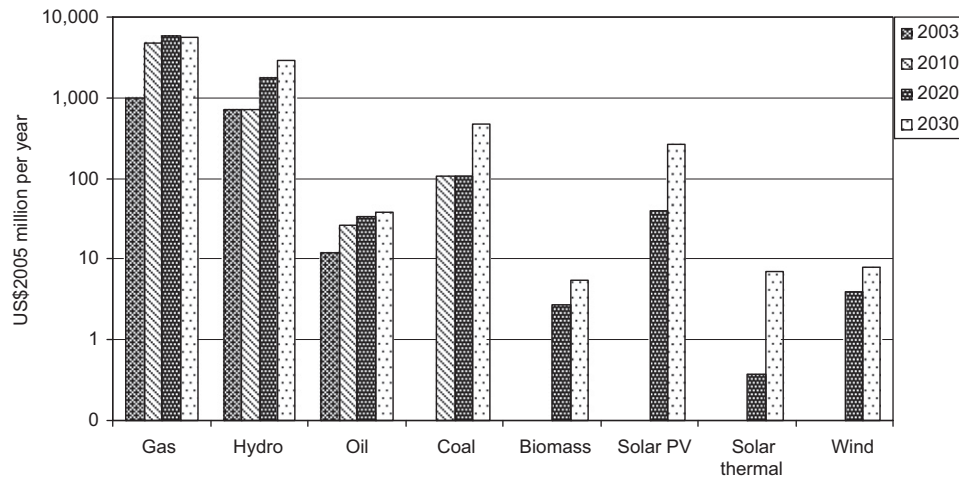


Fig. 18. Total costs by fuel for the electricity mix.

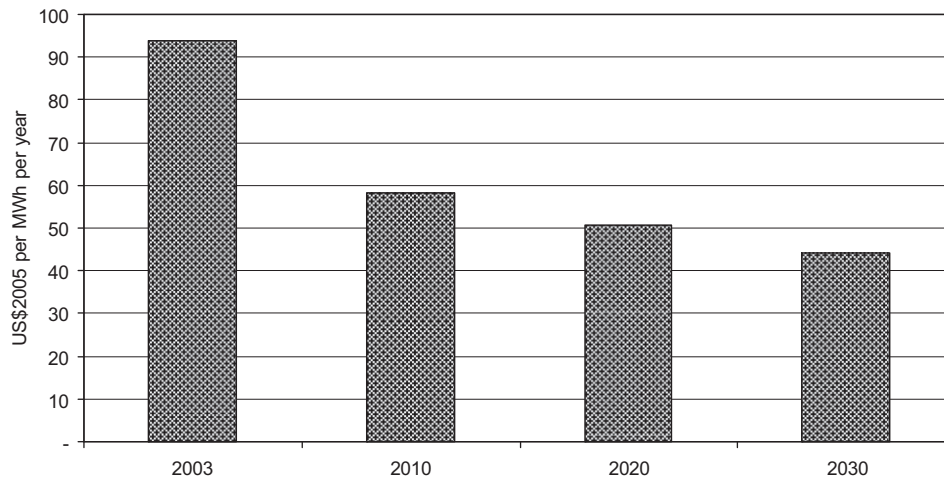


Fig. 19. Unit costs of the electricity mix.

provision of electricity using the conventional fuels would lead to much higher environmental impacts.

The concern over security of energy supply is one of the fundamental issues facing the Nigerian economy. It is argued within policy circles and energy analysts that increasing and expanding electricity supply from the grid would ensure stability of electricity supply to industries, businesses and residential sectors and would increase employment and productivity. Many firms and industries in Nigeria are under continued stress due to high production costs, and one of the principal factors that pushes up production costs is the high price of private power generation that firms tend to rely on due to the poor public provision of electricity (AfDB, 2009; Tyler, 2002). This has resulted in high manufacturing costs and reduced revenues for industry, often leading to factory closures and loss of jobs. Thus a stable electricity supply from the grid is likely to reduce manufacturing costs in industry and create stable conditions for employment. Moreover, expansion of the grid to the less remote rural areas through the Rural Electrification Fund (REF) also has the potential to extend and improve social benefits across the rural population that include improved education, healthcare, and national integration through access to media.

The planned expansion of the grid will also reduce the use of private generators (mainly diesel-fuelled) across the country. In addition to being expensive to run and therefore imposing heavy

financial burden on businesses and communities, these systems tend to be more environmentally damaging. These private generators are also a major source of noise pollution and given that they are mostly used at night, they present nuisance for neighbourhoods. Diesel engines also emit a complex mixture of air pollutants, composed of gaseous and solid material, which are responsible for a wide array of respiratory diseases, chronic obstructive lung disease, and lung cancer for non-smokers (Krivoshto et al., 2008). It should be noted that in cases where these systems are used within the household area, increased incidences of asthma attack are observed and death from the toxic fumes are also on increase in Nigeria (Reuters, 2008a, 2008b; This Day, 2008). Hence, the expansion of the grid would contribute to the phasing out of the private power generation sector, and could deliver real and tangible benefits in terms of improving wellbeing conditions.

Hydropower will account for about 50% of the total generation capacity by 2030; there are plans to add 10,388 MW of hydropower plants to the grid between 2010 and 2030, of which only 600 MW will be small hydropower systems (SHP). In addition to the low carbon energy generation that hydroelectricity represents, there will be additional benefits from the construction of these hydropower dams in the way of improvements in irrigation and water management. The social and economic gains could imply stabilising water and food security problems, which are

expected to become more acute in Nigeria with the increase in population and changes in lifestyles, especially in the urban areas.

Of course, the construction of large hydro (and irrigation) dams should not be taken lightly in places where traditional occupations and fishing practices are widespread. Dam construction has led to the displacement of rural and sometimes urban residents as well as altering local ecosystems in ways that often impose negative impacts on livelihoods, human settlement and cultures (WCD, 2000). Dam developments have been implicated in loss of vegetation and biodiversity. Thus, while the development of hydropower projects could be conceived with multiple benefits in mind such as directly increasing agricultural productivity and supplying clean drinking water to the rural areas, the design of such projects will need to integrate broader social and environmental considerations into the total cost of the project (WCD, 2000). This is likely to add significant costs to hydro projects in the future, although the costs are also project and site-dependent.

Despite the clear benefits of the proposed expansion of the power sector, the Government could still put more effort in addressing the sustainability concerns in this sector especially in the types of technologies being deployed in the future. More renewables should be introduced into the electricity mix as they have the potential to reduce the life cycle environmental impacts considerably, compared to the fossil fuels. Although renewable energy systems require significant investment costs, the total costs will even out over time due to little or no fuel costs involved with these systems, and the anticipated reduction in renewable energy technology prices, relative to conventional technologies and the anticipated increases in the prices of fossil-fuels (oil and gas) over time. However, the Government also faces a number of dilemmas in crafting policies that address multiple priorities in the provision of public service needs. For example, this is evidenced by the Government's concentrated efforts in meeting electricity demand and also addressing poverty reduction concerns in the near future, which may conflict with the need to invest in more sustainable energy options to meet the medium to long-term environmental imperatives.

Local/regional environmental impacts such as AP, EP and FTP increase by 50, 17 and 60 times, respectively, over the study period. The potential consequences from these impacts are likely to directly impact on local livelihoods, contributing to increased poverty in localised areas and potentially leading to social and political unrest. In Nigeria, most of the power plants and energy extraction activities are located in the Niger Delta region where ethnic and political unrests are already widespread, largely fuelled by discontent over environmental degradation linked to oil extraction (Human Rights Watch, 2005). The continuation of the 'business-as-usual' is likely to aggravate the fragile state of energy security in this region ranging from pipelines vandalism to kidnappings of oil workers (NNPC, 2006). Hence, the future of the energy sector and its expected growth would need to be justified on the basis of social and economic gains, not only for limited regions and social classes across Nigeria, but for a wider population base. Moreover, future policies and practices would need to be particularly sensitive to the development needs and aspirations of the oil-producing regions.

5. Conclusions

Energy security is one of the most urgent public policy issues being addressed by the Nigerian Government in order to meet its economic development objectives. The major focus has been to expand the electricity grid and exploit the many sources of energy within Nigeria to ensure a reliable and stable supply of electricity in the short, medium and long term. The Government has thus

focused on the economic and technical viability within its own means to develop its energy plans and policies. For example, all the power plants currently being built or approved for IPPs are fossil-fuelled systems consisting mainly of gas power plants. There are also concentrated efforts to utilise most of the hydropower potentials in the country. A major concern, therefore, is that the Government has not seriously considered and incorporated environmental (and social) issues in its plans. The analysis presented in this paper has been useful in highlighting the environmental and cost implications of the Government plans and also discussing the social implications of these plans.

The LCA results indicate a significant increase in the environmental impacts from the electricity sector over the period studied (2003–2030). Most of the life cycle environmental impacts occur due to the use of natural gas, oil (diesel) and coal in the energy mix. Renewable energy options such as solar, wind and hydro power do not require the burning of fuels for electricity generation and thus are the most favourable environmentally. Therefore improvements in the environmental performance of the electricity mix should focus on the increase in the use of renewables.

The economic analysis of the scenarios also shows that a significant investment is required to meet the objectives proposed by the Government with the total cost of the grid projected at over US\$9.40 billion per year by 2030, an over five-fold increase over the period. In terms of costs, the renewable energy options (solar, wind and biomass) are the least favourable electricity generation options. Thus achieving a sustainable electricity mix in Nigeria will require significant trade-off between economic costs and environmental benefits.

The modelling in this paper assumes a business-as-usual scenario where the composition of the electricity mix is based on the assumptions that the Government meets its target and the projected costs remain constant. However, this is not always the case as other probable policy directions in the future and changes in future costs could change the conclusions from this study. It is therefore necessary to conduct an exploratory study about other possible pathways with respect to the future electricity mix both in terms of the energy mix and energy costs. This will be presented in a follow-up to this paper.

Appendix A

Life cycle airborne inventory, environmental impacts and economic costs of the electricity mix are given in Tables A1–A3, respectively.

Table A1
Life cycle airborne inventory of the electricity mix.

Airborne emissions (tonnes)				
Substance	2003	2010	2020	2030
CO ₂	5.53E+06	5.47E+07	6.31E+07	7.18E+07
CO	1.87E+03	1.63E+04	2.03E+04	2.61E+04
CH ₄	5.10E+03	4.29E+04	5.00E+04	5.12E+04
Particulates	5.23E+02	5.76E+03	6.57E+03	1.95E+04
NMVOC	1.90E+03	1.03E+04	1.20E+04	1.35E+04
NO _x	4.52E+03	6.69E+04	7.66E+04	1.05E+05
N ₂ O	9.86E+01	9.35E+02	1.09E+03	1.12E+03
SO _x	3.07E+02	2.25E+04	2.32E+04	9.82E+04

Table A2

Life cycle environmental impacts of the electricity mix.

Substance	Type	2003	2010	2020	2030
GWP100 (tonne CO₂ eq.)					
Carbon dioxide	Air	5.53E+06	5.47E+07	6.31E+07	7.18E+07
Methane	Air	1.07E+05	9.87E+05	1.15E+06	1.17E+06
Carbon monoxide	Air	2.86E+03	2.49E+04	3.11E+04	3.99E+04
Nitrous oxide	Air	2.92E+04	2.77E+05	3.21E+05	3.31E+05
Others		2.71E+02	1.92E+04	4.30E+03	5.64E+04
Total GWP100		5.67E+06	5.60E+07	6.46E+07	7.34E+07
ADP (tonne Sb eq.)					
Natural gas	Fuel	4.95E+04	4.67E+05	5.43E+05	5.24E+05
Coal	Fuel	1.45E+03	3.72E+04	4.10E+04	1.44E+05
Crude oil	Fuel	4.89E+02	3.52E+03	4.22E+03	5.93E+03
Petroleum gas	Fuel	1.39E+02	5.67E+02	3.62E+02	4.88E+02
Total ADP		5.16E+04	5.09E+05	5.88E+05	6.74E+05
ODP (tonne CFC-11 eq.)					
Bromotrifluoromethane	Air	8.20E−02	5.90E−01	6.77E−01	9.59E−01
Tetrachloromethane	Air	2.11E−04	2.63E−03	6.00E−03	2.66E−02
Dichloroethane	Air	3.25E−08	5.94E−08	8.12E−06	1.10E−04
Total ODP		8.20E−02	5.92E−01	6.83E−01	9.85E−01
HTP (tonne 1,4 DB eq.)					
PAH	Air	4.32E+05	4.08E+06	4.75E+06	4.71E+06
Hydrogen fluoride	Air	6.17E+02	5.44E+05	5.46E+05	2.39E+06
Nitrogen oxides	Air	7.59E+03	8.03E+04	9.20E+04	1.26E+05
Nickel	Air	6.48E+03	4.79E+04	6.40E+04	1.26E+05
Selenium	Air	2.99E+03	4.69E+04	5.46E+04	1.64E+05
Arsenic	Air	1.52E+03	7.62E+04	8.27E+04	3.34E+05
Acenaphthylene	Water	9.81E+04	1.10E+06	1.38E+06	1.80E+06
Selenium	Water	4.57E+04	1.17E+06	1.40E+06	4.53E+06
Barium	Water	1.26E+04	2.46E+05	2.94E+05	8.66E+05
Barite	Water	1.13E+04	8.12E+04	9.73E+04	1.37E+05
PAH	Water	6.29E+03	4.54E+04	5.58E+04	8.59E+04
Vanadium	Water	2.63E+03	6.69E+04	7.91E+04	2.59E+05
Molybdenum	Water	2.25E+03	6.56E+04	6.88E+04	2.58E+05
Others		4.85E+03	8.96E+04	1.16E+05	3.14E+05
Total HTP		6.35E+05	7.74E+06	9.08E+06	1.61E+07
FTP (tonne 1,4 DB eq.)					
Vanadium	Water	7.46E+03	1.90E+05	2.25E+05	7.34E+05
Acenaphthylene	Water	9.68E+03	1.08E+05	1.36E+05	1.77E+05
Barium	Water	4.58E+03	8.92E+04	1.06E+05	3.13E+05
Nickel	Water	2.79E+03	6.91E+04	8.26E+04	2.66E+05
Barite	Water	4.08E+03	2.93E+04	3.51E+04	4.94E+04
Selenium	Water	2.38E+03	6.10E+04	7.29E+04	2.36E+05
Cobalt	Water	1.11E+03	2.85E+04	3.40E+04	1.10E+05
Copper	Water	9.81E+02	2.46E+04	2.94E+04	9.45E+04
PAH	Water	6.17E+02	4.46E+03	5.48E+03	8.44E+03
Zinc ion	Water	3.22E+02	5.44E+03	6.47E+03	1.69E+04
Molybdenum	Water	1.94E+02	5.67E+03	5.94E+03	2.23E+04
Others		8.35E+02	1.07E+04	1.21E+04	3.25E+04
Total FTP		3.50E+04	6.26E+05	7.51E+05	2.06E+06
MTP (tonne 1,4 DB eq.)					
Hydrogen fluoride	Air	8.82E+06	7.77E+09	7.80E+09	3.42E+10
Vanadium	Air	1.71E+06	1.63E+07	1.84E+07	4.36E+07
Selenium	Air	1.33E+06	2.08E+07	2.43E+07	7.28E+07
Nickel	Air	6.97E+05	5.15E+06	6.87E+06	1.36E+07
Beryllium	Air	1.80E+04	5.87E+06	5.90E+06	2.56E+07
Selenium	Water	2.07E+07	5.28E+08	6.31E+08	2.05E+09
Barium	Water	1.67E+07	3.26E+08	3.88E+08	1.14E+09
Barite	Water	1.49E+07	1.07E+08	1.29E+08	1.81E+08
Vanadium	Water	7.16E+06	1.82E+08	2.15E+08	7.02E+08
Nickel	Water	1.94E+06	4.80E+07	5.73E+07	1.85E+08
Cobalt	Water	1.43E+06	3.66E+07	4.37E+07	1.42E+08
Molybdenum	Water	8.51E+05	2.49E+07	2.61E+07	9.80E+07
Others		2.35E+05	9.38E+06	1.44E+07	4.64E+07
Total MTP		7.65E+07	9.08E+09	9.36E+09	3.89E+10
TTP (tonne 1,4 DB eq)					
Mercury	Air	6.97E+02	2.17E+04	2.28E+04	7.56E+04
Vanadium	Air	9.31E+01	8.86E+02	1.00E+03	2.38E+03
Nickel	Air	2.14E+01	1.59E+02	2.12E+02	4.18E+02
Arsenic	Air	7.03E+00	3.52E+02	3.82E+02	1.55E+03
Zinc	Air	6.35E+00	5.02E+01	7.78E+01	1.99E+02
Arsenic	Soil	1.79E+02	1.69E+03	1.97E+03	1.90E+03
Zinc	Soil	4.97E+01	4.68E+02	5.44E+02	5.26E+02
Others		1.91E+01	1.95E+02	3.14E+02	6.27E+02

Table A2 (continued)

Substance	Type	2003	2010	2020	2030
GWP100 (tonne CO₂ eq.)					
Total TTP		1.07E+03	2.55E+04	2.73E+04	8.32E+04
POCP (tonne C₂H₄ eq.)					
Sulphur oxides	Air	8.76E+00	1.08E+03	1.12E+03	4.71E+03
Ethane	Air	5.42E+01	5.11E+02	5.94E+02	5.74E+02
Carbon monoxide	Air	5.05E+01	4.39E+02	5.49E+02	7.04E+02
Pentane	Air	4.35E+01	4.10E+02	4.76E+02	4.70E+02
Butane	Air	3.87E+01	3.63E+02	4.22E+02	4.11E+02
Hexane	Air	3.61E+01	3.40E+02	3.95E+02	3.83E+02
Methane	Air	2.78E+01	2.57E+02	3.00E+02	3.07E+02
Propane	Air	2.37E+01	2.23E+02	2.59E+02	2.53E+02
Others		7.83E+00	7.67E+01	9.49E+01	1.48E+02
Total POCP		2.91E+02	3.70E+03	4.21E+03	7.96E+03
AP (tonne SO₂ eq.)					
Nitrogen oxides	Air	3.16E+03	3.34E+04	3.83E+04	5.24E+04
Sulphur oxides	Air	3.07E+02	2.76E+04	2.87E+04	1.19E+05
Ammonia	Air	1.17E+00	1.51E+01	2.95E+01	7.94E+01
Total AP		3.47E+03	6.11E+04	6.70E+04	1.71E+05
EP (tonne PO₄ eq.)					
Nitrogen oxides	Air	8.20E+02	8.69E+03	9.96E+03	1.36E+04
Ammonia	Air	2.56E+01	3.31E+00	6.45E+00	1.74E+01
Phosphorus, total	Air	6.54E+03	4.33E+00	4.35E+00	1.90E+01
Phosphate	Water	9.87E+00	2.51E+02	3.00E+02	9.72E+02
COD	Water	2.13E+00	1.98E+01	2.31E+01	2.29E+01
Phosphorus	Soil	2.06E+01	1.94E+02	2.25E+02	2.18E+02
Others		4.14E+00	1.76E+01	1.89E+01	5.07E+01
Total EP		8.57E+02	9.18E+03	1.05E+04	1.49E+04

Table A3

Economic costs of the electricity mix.

The capital and total costs of the electricity mix (US\$2005 million)				
Scenarios	2003	2010	2020	2030
Capital costs	9.42E+03	1.38E+04	2.18E+04	3.01E+04
Total cost (per year)	1.70E+03	5.61E+03	7.75E+03	9.40E+03
The various annualised costs contribution to the total costs of the electricity mix (US\$2005 million per year)				
Capital cost	1.24E+03	1.83E+03	2.86E+03	3.96E+03
Fixed cost	7.62E+01	1.56E+02	2.27E+02	3.03E+02
Variable cost	5.14E+01	2.72E+02	3.55E+02	3.99E+02
Fuel cost	3.33E+02	3.35E+03	4.30E+03	4.73E+03
Total cost	1.70E+03	5.61E+03	7.75E+03	9.40E+03
The total capital costs by fuel type in the electricity mix (US\$2005 million)				
Gas	4.13E+03	8.24E+03	8.45E+03	6.20E+03
Hydro	5.24E+03	5.24E+03	1.26E+04	2.06E+04
Oil	4.13E+01	4.13E+01	4.13E+01	4.13E+01
Coal	n/a	3.22E+02	3.22E+02	1.38E+03
Biomass	n/a	n/a	8.61E+00	1.72E+01
Solar PV	n/a	n/a	2.68E+02	1.78E+03
Solar thermal	n/a	n/a	2.31E+00	4.36E+01
Wind	n/a	n/a	2.39E+01	4.78E+01
The total costs by fuel type in the electricity mix (US\$2005 million per year)				
Gas	9.74E+02	4.76E+03	5.80E+03	5.72E+03
Hydro	7.15E+02	7.14E+02	1.76E+03	2.89E+03
Oil	1.18E+01	2.60E+01	3.33E+01	3.84E+01
Coal	n/a	1.08E+02	1.08E+02	4.74E+02
Biomass	n/a	n/a	2.68E+00	5.38E+00
Solar PV	n/a	n/a	3.89E+01	2.60E+02
Solar thermal	n/a	n/a	3.68E+01	7.08E+00
Wind	n/a	n/a	3.97E+00	7.94E+00
The 'unit costs of the electricity mix' (US\$2005/MWh.yr)				
Unit costs	9.40E+01	5.83E+01	5.06E+01	4.43E+01

References

- Adenikinju, A.F., 2003. Electric infrastructure failures in Nigeria: a survey-based analysis of the costs and adjustment responses. *Energy Policy* 31 (14), 1519–1530.
- Adenikinju, A.F., 2008. Efficiency of the energy sector and its impact on the competitiveness of the Nigerian economy international association of energy economics. Fourth Quarter 2008, pp. 27–31. Available at: <<http://www.iaee.org/documents/newsletterarticles/408adeola.pdf>>.
- African Development Bank (AfDB), 2009. Nigeria: Economic and Power Sector Reform Program (EPSERP). Appraisal Report. [Online]. Accessed: 20 February

2010. Available at: <http://www.afdb.org/fileadmin/uploads/afdb/Documents/Project-and-Operations/Nigeria%20Energy%20sector%20Reform%20Program.pdf>.
- Akarakiri, J.B., 1999. Private electric power generation as an alternative in Nigeria. *Energy* 24 (5), 445–447.
- Akinbami, J.F., 2001. Renewable energy resources and technologies in Nigeria: present situation, future prospects and policy framework. *Mitigation and Adaptation Strategies for Global Change* 6, 155–181.
- Amioku, T.O., 2002. Developing Power and Energy Projects in Delta State. [Online]. Accessed: 21 March 2004. Available at: <http://www.deltastate.gov.ng/petroenergy.htm>.
- Babatunde, M.A., Shuaibu, M.I., 2009. The demand for residential electricity in Nigeria: a bound testing approach. In: The Conference Paper, Presented at the Second International Workshop on Empirical Methods in Energy Economics, University of Alberta, Canada. 28–29 August 2009. Available at: www.africametrics.org/documents/conference09/.../Babatunde_Shuaibu.pdf.
- Banerjee, S., Duckers, L.J., Blanchard, R., Choudhury, B.K., 2006. Life cycle analysis of selected solar and wave energy systems. *Advances in Energy Research*. [Online]. Accessed: 12 January 2007. Available at: http://www.esi.iitb.ac.in/aer2006_files/papers/142.pdf.
- Central Bank of Nigeria (CBN), 2003. Economic Trends. [Online]. Accessed: 4 July 2004. Available at: <http://www.cenbank.org>.
- Central Bank of Nigeria (CBN), 2006. Economic Report. [Online]. Accessed: 12 June 2007. Available at: <http://www.cenbank.org>.
- Dones, R., Zhou, X., Tian, C., 2004. Life cycle assessment (LCA) of Chinese energy chains for Shandong electricity scenarios. *International Journal of Global Energy Issues* 22, 199–224.
- Energy Commission of Nigeria (ECN), 2003. National Energy Policy. Document Presented to the Presidency. [Online]. Accessed: 09 December 2008. Available at: http://www.energy.gov.ng/index.php?option=com_docman&task=cat_view&Itemid=&gid=19&orderby=dmdate_published&asc=DESC.
- Energy Commission of Nigeria (ECN), 2004. Energy Master Plan Study for Nigeria. On-going Project with the International Atomic Energy Agency (IAEA).
- Energy Commission of Nigeria (ECN), United Nations Development Programme (UNDP), 2005. Renewable Energy Master Plan. Document presented to the Federal Government.
- Energy Information Administration (EIA), 2003. Country Analysis Briefs: Nigeria. [Online]. Accessed: 12 January 2004. Available at: <http://www.eia.doe.gov/emeu/cabs/nigeria.html>.
- Energy Information Administration (EIA), 2005a. Country Analysis Briefs: Nigeria. [Online]. Accessed: 09 December 2006. Available at: <http://www.eia.doe.gov>.
- Energy Information Administration (EIA), 2005b. Electricity Market Modules. [Online]. Accessed: 30 January 2005. Available at: [http://www.eia.doe.gov/oiaf/archive/aeo05/assumption/pdf/0554\(2005\).pdf](http://www.eia.doe.gov/oiaf/archive/aeo05/assumption/pdf/0554(2005).pdf).
- Energy Information Administration (EIA), 2006. Energy Forecasts and Analysis. [Online]. Accessed: 11 May 2007. Available at: <http://www.eia.doe.gov/oiaf/forecasting.html>.
- Energy Information Administration (EIA), 2007a. Electricity Market Modules. [Online]. Accessed: 11 December 2009. Available at: <http://www.eia.doe.gov/oiaf/archive/aeo07/assumption/pdf/electricity.pdf>.
- Energy Information Administration (EIA), 2007b. Renewable Fuels Module. [Online]. Accessed: 11 December 2009. Available at: <http://www.eia.doe.gov/oiaf/archive/aeo07/assumption/pdf/renewable.pdf>.
- Environmental Rights Action (ERA), 2005. Gas Flaring in Nigeria: A Human Right's, Environmental and Economic Monstrosity. [Online]. Accessed: 15 January 2008. Available at: http://www.foe.co.uk/resource/reports/gas_flaring_nigeria.pdf.
- Gorokhov, V., Manfredo, L., Ramezan, M., Ratafia-Brown, J., Stiegel, G., 2002. Life cycle analysis of advanced power generation systems. *Technology* 8, 217–228.
- Guinée, J.B., Gorée, M., Heijungs, R., Huppes, G., Kleijn, R., van Oers, L., Wegener Sleeswijk, A., Suh, S., Udo de Haes, H.A., de Bruijn, H., van Duin, R., Huijbregts, M.A.J., 2001. Life Cycle Assessment: An Operational Guide to the ISO Standards. Parts 1, 2a & 2b. Kluwer Academic Publishers, Dordrecht, The Netherlands.
- Hondo, H., 2004. Life cycle GHG emission analysis of power generation systems: Japanese case. *Energy* 30 (2005), 2042–2056.
- Human Rights Watch, 2005. Rivers and Blood: Guns, Oil and Power in Nigeria's Rivers State. A Human Rights Watch Briefing Paper. [Online]. Accessed: 13 June 2009. Available at: <http://www.hrw.org/legacy/background/africa/nigeria0205/nigeria0205.pdf>.
- Ibitoye, F.I., Adenikinju, A., 2007. Future demand for electricity in Nigeria. *Applied Energy* 84, 492–504.
- Ikeme, J., Ebohon, O.J., 2005. Nigeria's electric power sector reform: What should be the key objectives? *Energy Policy* 33 (9) 1213–1221.
- Kannan, R., Leong, K.C., Osman, R., Ho, H.K., 2006. Life cycle energy, emissions and cost inventory of power generation technologies in Singapore. *Renewable and Sustainable Energy Reviews* 11 (2007), 702–715.
- Koskela, S., Seppala, J., Lipp, A., Hiltune, M., Pold, E., Talve, S., 2007. Estonia electricity supply scenarios for 2020 and their environmental performance. *Energy Policy* 35 (2007), 3571–3582.
- Krivoshio, I.N., Richards, J.R., Albertson, T.E., Derlet, R.W., 2008. The toxicity of diesel exhaust: Implications for primary care. *The Journal of the American Board of Family Medicine* 21 (1), 55–62.
- Mbendi Information Services, 2005. Electrical Power in Nigeria—Overview. [Online]. Accessed: 12 January 2010. Available at: <http://www.mbendi.com/indy/powr/af/ng/p0005.htm>.
- Meier, P.J., Wilson, P.H., Kulcinski, G.L., Denholm, P.I., 2005. US industry response to carbon constraints: a life-cycle assessment of supply alternatives. *Energy Policy* 33, 1099–1108.
- National Electric Power Authority (NEPA), 2000. Infrastructure: Generation Stations. Accessed: 09 July 2004. Available at: <http://www.nepanigeria.org/structure.html>.
- National Electric Power Authority (NEPA), 2004. Electricity Generation Statistics. Accessed: 13 July 2004. Available at: <http://www.nepanigeria.org/structure.html>.
- Nigeria Coal Corporation (NCC), 2002. Reviving the Coal Industry. Abuja, Nigeria.
- Nigerian National Petroleum Corporation (NNPC), 2004. Annual Statistical Bulletin 2003. [Online]. Accessed: 13 August 2004. Available at: www.nnpc-nigeria.com.
- Nigerian National Petroleum Corporation (NNPC), 2005. Nigeria Gas 2005. [Online]. Accessed: 15 November 2009. Available at: <http://www.nnpcgroup.com/nigeria-gas>.
- Nigerian National Petroleum Corporation (NNPC), 2006. Annual Statistical Bulletin 2005. [Online]. Accessed: 10 June, 2006. Available at: www.nnpc-nigeria.com.
- Obadote, D.J., 2009. Energy crisis in Nigeria: Technical issues and solutions. In: The Conference Paper, Presented at Power Sector Prayer Conference, Nigeria. June 25–27, 2009.
- Öko Institute, 2006. Global Emission Model for Integrated Systems (GEMIS) Version 4.3. Germany Available at: <http://www.oeko.de/service/gemis/en/index.htm>.
- Okpanefe, P.E., Owolabi, S., 2002. Small hydropower in Nigeria. SHP News, Spring 2002. TCDC Training Workshop on SHP, 2001. Available at: <http://www.hrcshp.org/shp/en/pdf/2002/springdetail.pdf>.
- Organisation of Petroleum Exporting Countries (OPEC), 2003. Average Prices of Petroleum. [Online]. Accessed: 20 December 2005. Available at: <http://www.opec.org>.
- Pehnt, M., 2003. Assessing future energy and transport systems: the case of fuel cells. Part I: methodological aspects. *International Journal of LCA* 8 LCA 5, 283–289.
- Pehnt, M., 2006. Dynamic life cycle assessment (LCA) of renewable energy technologies. *Renewable Energy* 31, 55–71.
- Pieu, N., 2007. Life cycle inventory study of the electrical energy production in Romania. *International Journal of LCA* 12 (4), 225–229.
- Reuters, 2008a. Generator Fumes Kill Family of 13 in Nigeria. [Online]. Accessed: 17 December 2008. Available at: <http://www.alertnet.org/thenews/newsdesk/L6129732.htm>.
- Reuters, 2008b. Generator Fumes kill 17 at Nigeria Prayer Meeting. [Online]. Accessed: 11 December 2009. Available at: <http://www.reuters.com/article/idUSL16498503>.
- Ruether, J.A., Ramezan, M., Balash, P.C., 2004. Greenhouse gas emissions from coal gasification power generation systems. *Journal of Infrastructure Systems* 10 (3), 111–119.
- Sambo, A.S., 2009. Strategic developments in renewable energy in Nigeria. *International Association of Energy Economics*. Third Quarter 2009. pp. 15–19. Available at: www.iaee.org/en/publications/newsletterdl.aspx?id=75.
- Sonibare, J.A., Akeredolu, F.A., 2006. Natural gas domestic market development for total elimination of routine flares in Nigeria's upstream petroleum operations. *Energy Policy* 34, 743–753.
- Tallapragada, P.V.S.N., 2009. Nigeria's electricity sector—electricity and gas pricing barriers. *International Association of Energy economics*. First Quarter 2009. pp. 29–34. Available at: <http://www.iaee.org/en/publications/newsletterdl.aspx?id=72>.
- The World Bank Group (WBG), 2007. Nigeria Data Profile. [Online]. Accessed: 12 March, 2007. Available at: <http://devdata.worldbank.org/external/CPProfile.asp?CCODE=NGA&PTYPE=CP>.
- This Day, 2008. Generator fumes kill Four. Publication date: 04 December 2008. [Online]. Accessed: 11 December 2009. Available at: <http://allafrica.com/stories/200812040314.html>.
- Tyler, G., 2002. Nigeria: Public and Private Electricity Provision as a Barrier to Manufacturing Competitiveness. Findings Report—221. The World Bank Group, Washington, DC.
- Ukpohor, E.T.O., 2009. Nigerian gas master plan: strengthening the Nigeria Gas Infrastructure Blueprint as a base for expanding regional gas market. Technical Paper for World Gas Conference (WGC), 2009. Available at: <http://www.igu.org/html/wgc2009/papers/docs/wgcFinal00764.pdf>.
- World Commission of Dams (WCD), 2000. Dams and Development—a new framework for decision-making. [Online]. Accessed: 13 May 2009. Available at: <http://www.dams.org/commission/publications.htm>.
- World Energy Council (WEC), 2004. Comparison of Energy Systems Using Life Cycle Assessment. [Online]. Accessed: 12 January 2007. Available at: <http://www.worldenergy.org/documents/lca2.pdf>.